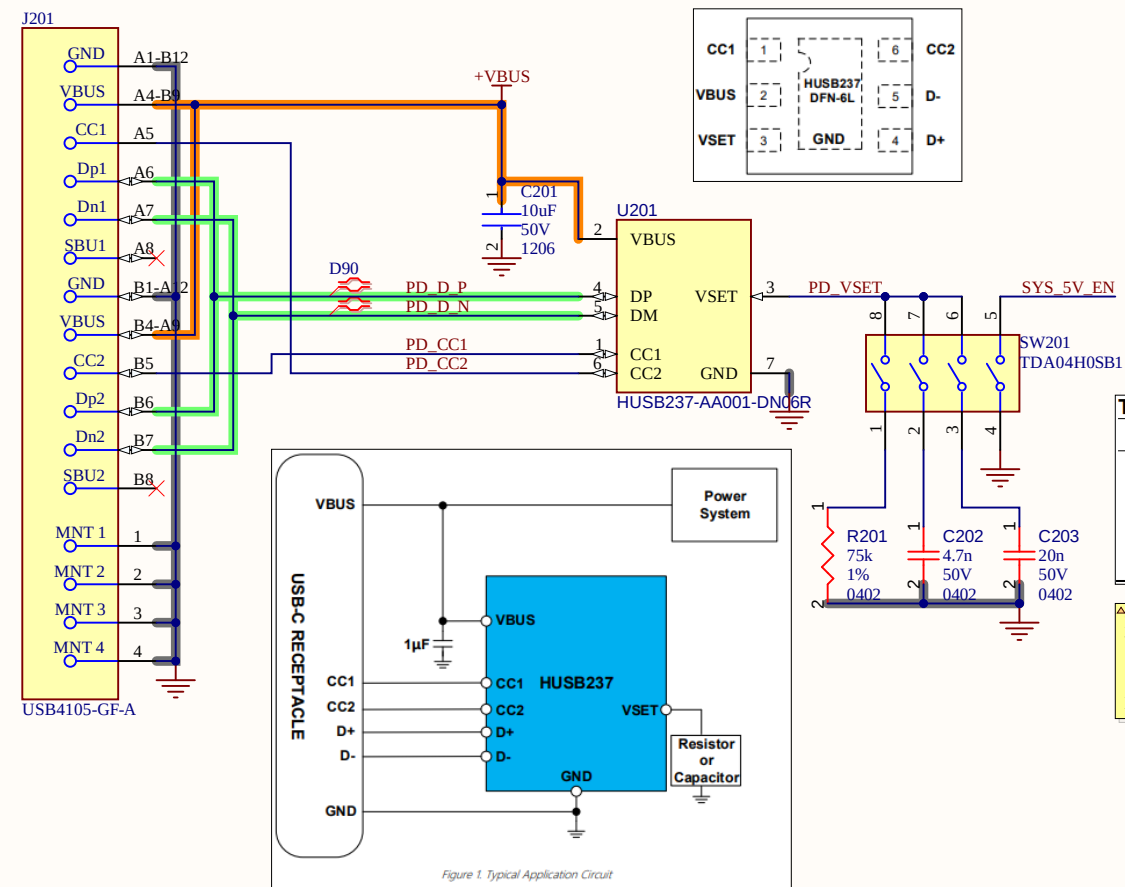




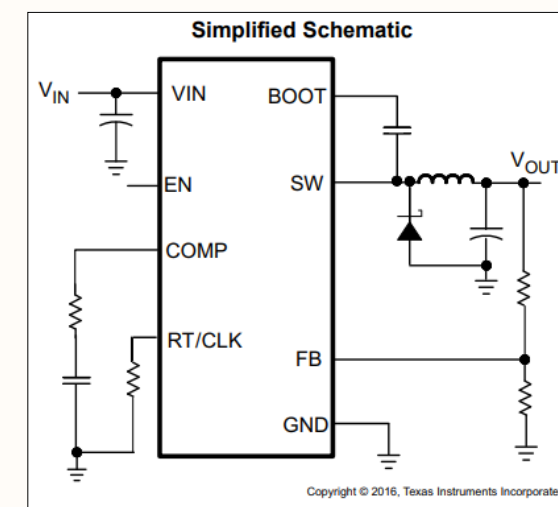
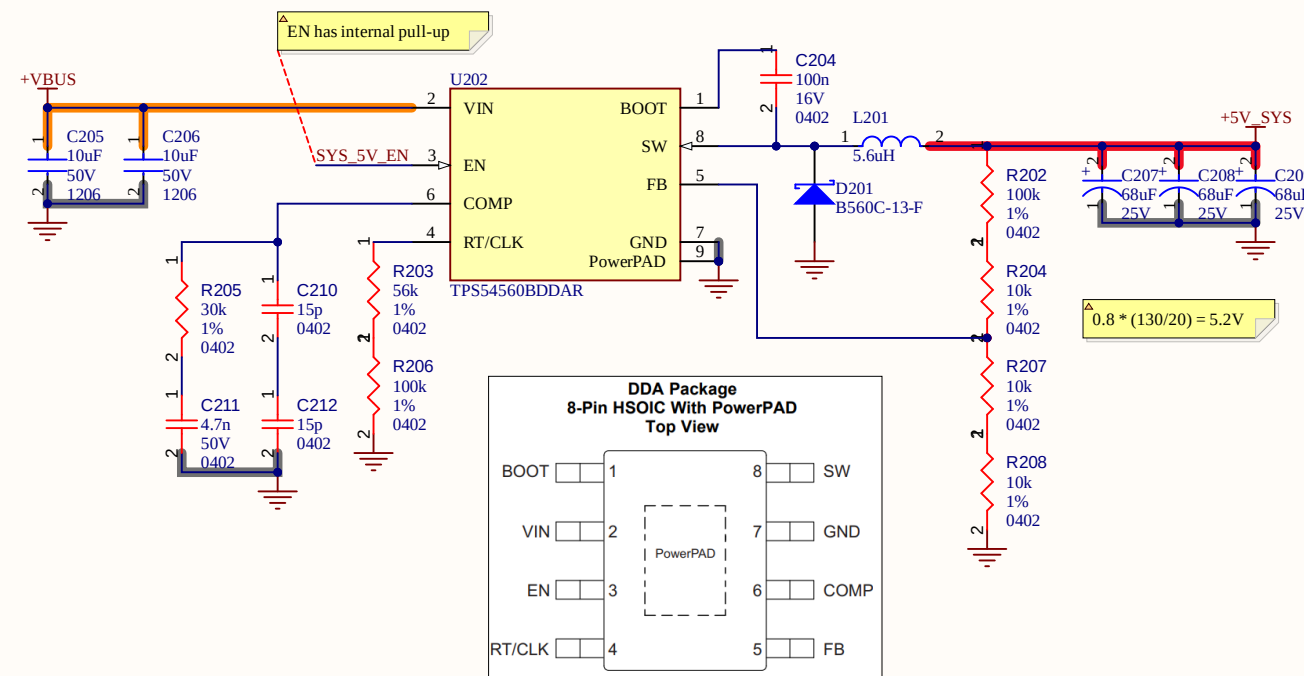


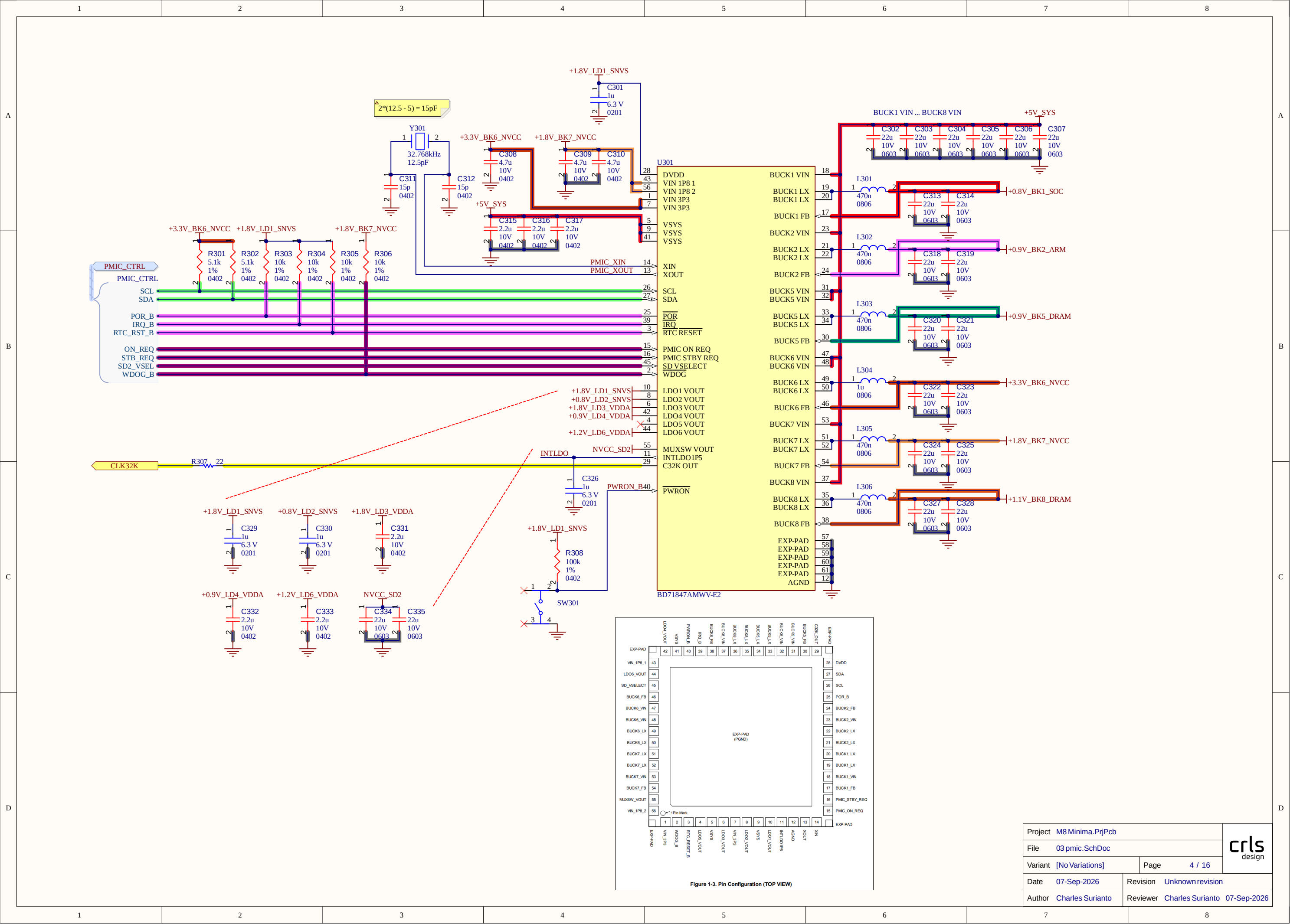


| VSET Pin Configuration | Preferred Voltage (V) |
|------------------------|-----------------------|
| 0 $\Omega$             | 5                     |
| 75 k $\Omega$          | 9                     |
| open                   | 12                    |
| 4.7 nF                 | 15                    |
| 20 nF                  | 20                    |

△ DS: Connect a resistor or capacitor with 1% tolerance between VSET and GND to indicate the target RDO voltage value.

\*10% cap should suffice since the difference is a factor of 5, provided DC bias is accounted for.









| Ball name  | LPDDR4  |
|------------|---------|
| DRAM_DQ0_P | DQ0_1_A |
| DRAM_DQ0_N | DQ0_1_B |
| DRAM_DM0   | DM0_A   |
| DRAM_DQ00  | DQ0_A   |
| DRAM_DQ01  | DQ1_A   |
| DRAM_DQ02  | DQ2_A   |
| DRAM_DQ03  | DQ3_A   |
| DRAM_DQ04  | DQ4_A   |
| DRAM_DQ05  | DQ5_A   |

|             |          |
|-------------|----------|
| DRAM_DQ06   | DQ6_A    |
| DRAM_DQ07   | DQ7_A    |
| DRAM_DQ08_P | DQ08_1_A |
| DRAM_DQ08_N | DQ08_1_B |
| DRAM_DM1    | DM1_A    |
| DRAM_DQ08   | DQ08_A   |
| DRAM_DQ09   | DQ09_A   |
| DRAM_DQ10   | DQ10_A   |
| DRAM_DQ11   | DQ11_A   |
| DRAM_DQ12   | DQ12_A   |
| DRAM_DQ13   | DQ13_A   |
| DRAM_DQ14   | DQ14_A   |
| DRAM_DQ15   | DQ15_A   |
| DRAM_DQ02_P | DQ02_1_B |
| DRAM_DQ02_N | DQ02_1_A |
| DRAM_DM2    | DM2_B    |
| DRAM_DQ16   | DQ16_B   |
| DRAM_DQ17   | DQ17_B   |
| DRAM_DQ18   | DQ18_B   |
| DRAM_DQ19   | DQ19_B   |
| DRAM_DQ20   | DQ20_B   |
| DRAM_DQ21   | DQ21_B   |
| DRAM_DQ22   | DQ22_B   |
| DRAM_DQ23   | DQ23_B   |
| DRAM_DQ03_P | DQ03_1_B |
| DRAM_DQ03_N | DQ03_1_A |
| DRAM_DM3    | DM3_B    |
| DRAM_DQ24   | DQ24_B   |
| DRAM_DQ25   | DQ25_B   |
| DRAM_DQ26   | DQ26_B   |
| DRAM_DQ27   | DQ27_B   |
| DRAM_DQ28   | DQ28_B   |
| DRAM_DQ29   | DQ29_B   |

|              |         |
|--------------|---------|
| DRAM_DQ30    | DQ30_B  |
| DRAM_DQ31    | DQ31_B  |
| DRAM_RESET_N | RESET_N |
| DRAM_ALERT_N | MTEST1  |
| DRAM_AC00    | CKE0_A  |
| DRAM_AC01    | CKE1_A  |
| DRAM_AC02    | CS0_A   |
| DRAM_AC03    | CS1_A   |
| DRAM_AC04    | CK_1_A  |
| DRAM_AC05    | CK_c_A  |
| DRAM_AC06    | ---     |
| DRAM_AC07    | ---     |
| DRAM_AC08    | CA0_A   |
| DRAM_AC09    | CA1_A   |
| DRAM_AC10    | CA2_A   |
| DRAM_AC11    | CA3_A   |
| DRAM_AC12    | CA4_A   |
| DRAM_AC13    | CA5_A   |
| DRAM_AC14    | ---     |
| DRAM_AC15    | ---     |
| DRAM_AC16    | ---     |
| DRAM_AC17    | ---     |
| DRAM_AC19    | MTEST   |
| DRAM_AC20    | CKE0_B  |
| DRAM_AC21    | CKE1_B  |
| DRAM_AC22    | CS1_B   |
| DRAM_AC23    | CS0_B   |
| DRAM_AC24    | CK_1_B  |
| DRAM_AC25    | CK_c_B  |
| DRAM_AC26    | ---     |
| DRAM_AC27    | ---     |
| DRAM_AC28    | CA0_B   |
| DRAM_AC29    | CA1_B   |
| DRAM_AC30    | CA2_B   |

|           |       |
|-----------|-------|
| DRAM_AC31 | CA3_B |
| DRAM_AC32 | CA4_B |
| DRAM_AC33 | CA5_B |
| DRAM_AC34 | ---   |
| DRAM_AC35 | ---   |
| DRAM_AC36 | ---   |
| DRAM_AC37 | ---   |
| DRAM_AC38 | ---   |
| DRAM_ZN   | ZQ    |
| DRAM_VREF | VREF  |

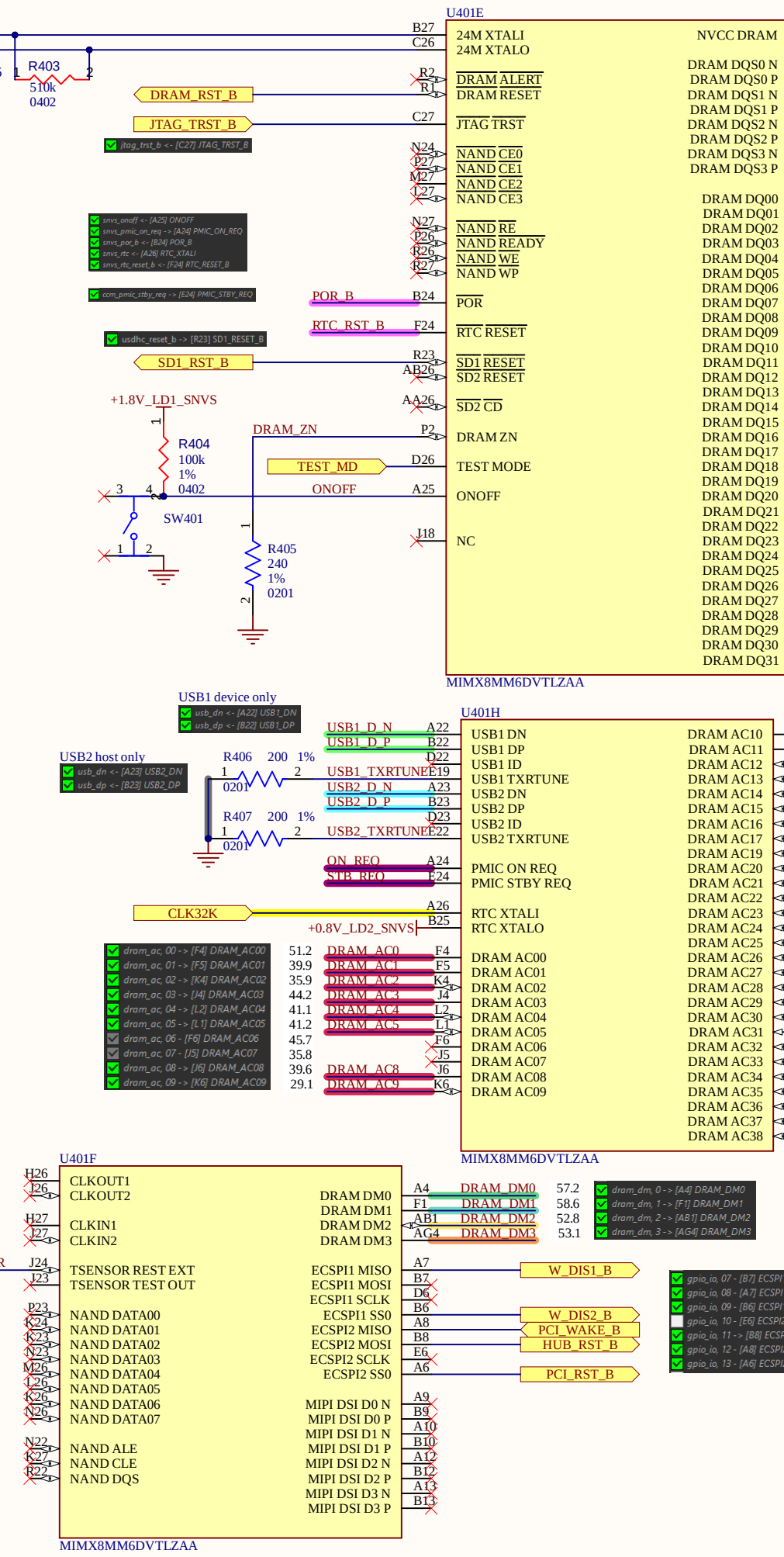
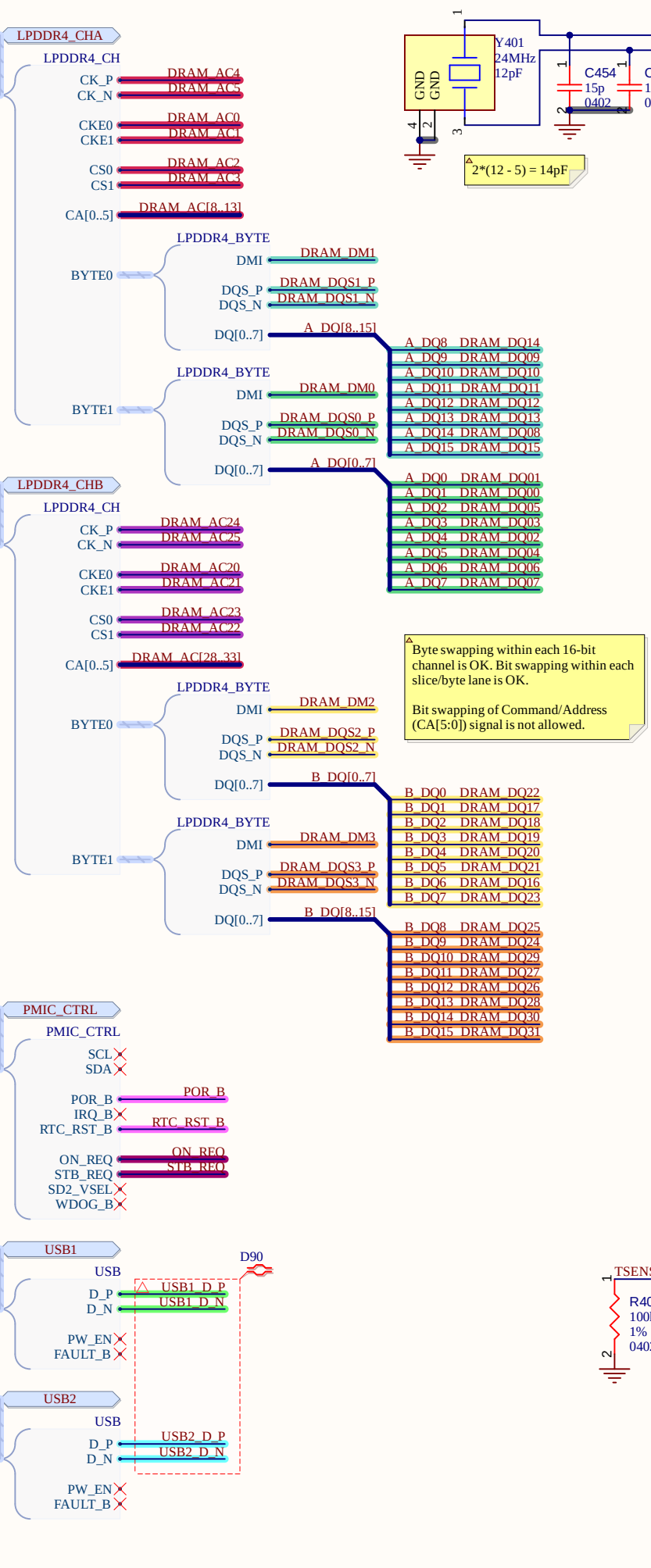
|              |         |
|--------------|---------|
| DRAM_DQ30    | DQ14_B  |
| DRAM_DQ31    | DQ15_B  |
| DRAM_RESET_N | RESET_N |
| DRAM_ALERT_N | MTEST1  |
| DRAM_AC00    | CKE0_A  |
| DRAM_AC01    | CKE1_A  |
| DRAM_AC02    | CS0_A   |
| DRAM_AC03    | CS1_A   |
| DRAM_AC04    | CK_1_A  |
| DRAM_AC05    | CK_c_A  |
| DRAM_AC06    | ---     |
| DRAM_AC07    | ---     |
| DRAM_AC08    | CA0_A   |
| DRAM_AC09    | CA1_A   |
| DRAM_AC10    | CA2_A   |
| DRAM_AC11    | CA3_A   |
| DRAM_AC12    | CA4_A   |
| DRAM_AC13    | CA5_A   |
| DRAM_AC14    | ---     |
| DRAM_AC15    | ---     |
| DRAM_AC16    | ---     |
| DRAM_AC17    | ---     |
| DRAM_AC19    | MTEST   |
| DRAM_AC20    | CKE0_B  |
| DRAM_AC21    | CKE1_B  |
| DRAM_AC22    | CS1_B   |
| DRAM_AC23    | CS0_B   |
| DRAM_AC24    | CK_1_B  |
| DRAM_AC25    | CK_c_B  |
| DRAM_AC26    | ---     |
| DRAM_AC27    | ---     |
| DRAM_AC28    | CA0_B   |
| DRAM_AC29    | CA1_B   |
| DRAM_AC30    | CA2_B   |

|           |       |
|-----------|-------|
| DRAM_AC31 | CA3_B |
| DRAM_AC32 | CA4_B |
| DRAM_AC33 | CA5_B |
| DRAM_AC34 | ---   |
| DRAM_AC35 | ---   |
| DRAM_AC36 | ---   |
| DRAM_AC37 | ---   |
| DRAM_AC38 | ---   |
| DRAM_ZN   | ZQ    |
| DRAM_VREF | VREF  |

|           |       |
|-----------|-------|
| DRAM_AC31 | CA3_B |
| DRAM_AC32 | CA4_B |
| DRAM_AC33 | CA5_B |
| DRAM_AC34 | ---   |
| DRAM_AC35 | ---   |
| DRAM_AC36 | ---   |
| DRAM_AC37 | ---   |
| DRAM_AC38 | ---   |
| DRAM_ZN   | ZQ    |
| DRAM_VREF | VREF  |

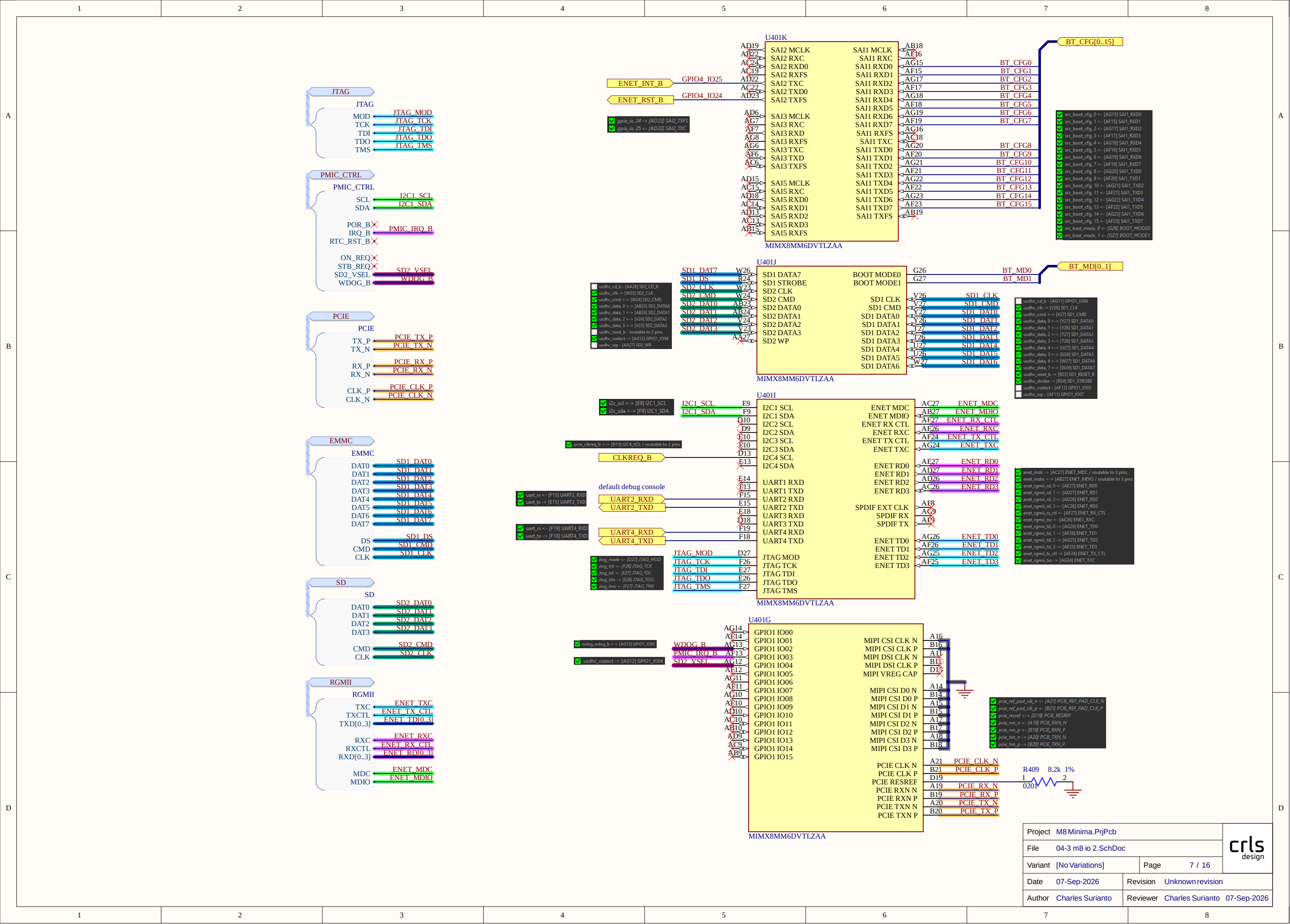
|           |       |
|-----------|-------|
| DRAM_AC31 | CA3_B |
| DRAM_AC32 | CA4_B |
| DRAM_AC33 | CA5_B |
| DRAM_AC34 | ---   |
| DRAM_AC35 | ---   |
| DRAM_AC36 | ---   |
| DRAM_AC37 | ---   |
| DRAM_AC38 | ---   |
| DRAM_ZN   | ZQ    |
| DRAM_VREF | VREF  |

|           |       |
|-----------|-------|
| DRAM_AC31 | CA3_B |
| DRAM_AC32 | CA4_B |
| DRAM_AC33 | CA5_B |
| DRAM_AC34 | ---   |
| DRAM_AC35 | ---   |
| DRAM_AC36 | ---   |
| DRAM_AC37 | ---   |
| DRAM_AC38 | ---   |
| DRAM_ZN   | ZQ    |
| DRAM_VREF | VREF  |



|                                                   |             |      |                                     |
|---------------------------------------------------|-------------|------|-------------------------------------|
| <div><div></div><div>+1.1V_BK8_DRAM</div></div>   |             |      |                                     |
| A2                                                | DRAM_DQS0_N | 58.9 | ✓ dram_dqs0.n <-> [B2] DRAM_DQS0_N  |
| A2                                                | DRAM_DQS0_P | 59   | ✓ dram_dqs0.p <-> [A2] DRAM_DQS0_P  |
| A1                                                | DRAM_DQS1_N | 47.2 | ✓ dram_dqs1.n <-> [H1] DRAM_DQS1_N  |
| A1                                                | DRAM_DQS1_P | 48.6 | ✓ dram_dqs1.p <-> [G1] DRAM_DQS1_P  |
| A1                                                | DRAM_DQS2_N | 48.6 | ✓ dram_dqs2.n <-> [Y1] DRAM_DQS2_N  |
| A1                                                | DRAM_DQS2_P | 49.9 | ✓ dram_dqs2.p <-> [AA1] DRAM_DQS2_P |
| A2                                                | DRAM_DQS3_N | 55   | ✓ dram_dqs3.n <-> [AF2] DRAM_DQS3_N |
| A2                                                | DRAM_DQS3_P | 55.5 | ✓ dram_dqs3.p <-> [AG2] DRAM_DQS3_P |
| A5                                                | DRAM_DQ00   | 47.2 | ✓ dram_dq_00 <-> [A5] DRAM_DQ00     |
| A5                                                | DRAM_DQ01   | 43   | ✓ dram_dq_01 <-> [B5] DRAM_DQ01     |
| A2                                                | DRAM_DQ02   | 54.6 | ✓ dram_dq_02 <-> [D2] DRAM_DQ02     |
| A1                                                | DRAM_DQ03   | 51.7 | ✓ dram_dq_03 <-> [D1] DRAM_DQ03     |
| A1                                                | DRAM_DQ04   | 59.9 | ✓ dram_dq_04 <-> [C1] DRAM_DQ04     |
| A1                                                | DRAM_DQ05   | 58.1 | ✓ dram_dq_05 <-> [B1] DRAM_DQ05     |
| A3                                                | DRAM_DQ06   | 64.6 | ✓ dram_dq_06 <-> [A3] DRAM_DQ06     |
| A4                                                | DRAM_DQ07   | 51.4 | ✓ dram_dq_07 <-> [B4] DRAM_DQ07     |
| A2                                                | DRAM_DQ08   | 45   | ✓ dram_dq_08 <-> [F2] DRAM_DQ08     |
| A2                                                | DRAM_DQ09   | 50.1 | ✓ dram_dq_09 <-> [G2] DRAM_DQ09     |
| A1                                                | DRAM_DQ10   | 46.2 | ✓ dram_dq_10 <-> [J1] DRAM_DQ10     |
| A2                                                | DRAM_DQ11   | 47.2 | ✓ dram_dq_11 <-> [J2] DRAM_DQ11     |
| A2                                                | DRAM_DQ12   | 40.3 | ✓ dram_dq_12 <-> [K2] DRAM_DQ12     |
| A1                                                | DRAM_DQ13   | 48.8 | ✓ dram_dq_13 <-> [K1] DRAM_DQ13     |
| A1                                                | DRAM_DQ14   | 58.4 | ✓ dram_dq_14 <-> [E1] DRAM_DQ14     |
| A2                                                | DRAM_DQ15   | 52.4 | ✓ dram_dq_15 <-> [F1] DRAM_DQ15     |
| A2                                                | DRAM_DQ16   | 51.4 | ✓ dram_dq_16 <-> [AB2] DRAM_DQ16    |
| A2                                                | DRAM_DQ17   | 49.9 | ✓ dram_dq_17 <-> [AA2] DRAM_DQ17    |
| A1                                                | DRAM_DQ18   | 54.5 | ✓ dram_dq_18 <-> [W1] DRAM_DQ18     |
| A2                                                | DRAM_DQ19   | 42   | ✓ dram_dq_19 <-> [W2] DRAM_DQ19     |
| A2                                                | DRAM_DQ20   | 53.6 | ✓ dram_dq_20 <-> [V2] DRAM_DQ20     |
| A1                                                | DRAM_DQ21   | 49.8 | ✓ dram_dq_21 <-> [V1] DRAM_DQ21     |
| A1                                                | DRAM_DQ22   | 54.7 | ✓ dram_dq_22 <-> [AC1] DRAM_DQ22    |
| A2                                                | DRAM_DQ23   | 48   | ✓ dram_dq_23 <-> [AC2] DRAM_DQ23    |
| A5                                                | DRAM_DQ24   | 60.2 | ✓ dram_dq_24 <-> [AG5] DRAM_DQ24    |
| A5                                                | DRAM_DQ25   | 51.3 | ✓ dram_dq_25 <-> [E2] DRAM_DQ25     |
| A2                                                | DRAM_DQ26   | 48.8 | ✓ dram_dq_26 <-> [AD2] DRAM_DQ26    |
| D1                                                | DRAM_DQ27   | 58   | ✓ dram_dq_27 <-> [AD1] DRAM_DQ27    |
| A1                                                | DRAM_DQ28   | 59.1 | ✓ dram_dq_28 <-> [AE1] DRAM_DQ28    |
| A1                                                | DRAM_DQ29   | 58.1 | ✓ dram_dq_29 <-> [AF1] DRAM_DQ29    |
| A3                                                | DRAM_DQ30   | 50   | ✓ dram_dq_30 <-> [AG3] DRAM_DQ30    |
| A4                                                | DRAM_DQ31   | 44.2 | ✓ dram_dq_31 <-> [AF4] DRAM_DQ31    |
| <div><div>A</div><div>Pin delay in ps</div></div> |             |      |                                     |
| A4                                                | DRAM_AC10   | 54.8 | ✓ dram_ac_10-> [E4] DRAM_AC10       |
| A4                                                | DRAM_AC11   | 59.7 | ✓ dram_ac_11-> [D5] DRAM_AC11       |
| A4                                                | DRAM_AC12   | 33.8 | ✓ dram_ac_12-> [N4] DRAM_AC12       |
| A4                                                | DRAM_AC13   | 32   | ✓ dram_ac_13-> [N5] DRAM_AC13       |
| A4                                                | DRAM_AC14   | 43.1 | ✓ dram_ac_14-> [K5] DRAM_AC14       |
| A4                                                | DRAM_AC15   | 22.4 | ✓ dram_ac_15-> [N6] DRAM_AC15       |
| A4                                                | DRAM_AC16   | 39.1 | ✓ dram_ac_16-> [M1] DRAM_AC16       |
| A4                                                | DRAM_AC17   | 39.4 | ✓ dram_ac_17-> [M2] DRAM_AC17       |
| A4                                                | DRAM_AC19   | 43.4 | ✓ dram_ac_19-> [N2] DRAM_AC19       |
| A4                                                | DRAM_AC20   | 51.6 | ✓ dram_ac_20-> [AB4] DRAM_AC20      |
| A5                                                | DRAM_AC21   | 51.6 | ✓ dram_ac_21-> [AB5] DRAM_AC21      |
| A4                                                | DRAM_AC22   | 47.7 | ✓ dram_ac_22-> [W4] DRAM_AC22       |
| A4                                                | DRAM_AC23   | 40   | ✓ dram_ac_23-> [J4] DRAM_AC23       |
| A4                                                | DRAM_AC24   | 41.8 | ✓ dram_ac_24-> [U2] DRAM_AC24       |
| A4                                                | DRAM_AC25   | 42.2 | ✓ dram_ac_25-> [U1] DRAM_AC25       |
| A4                                                | DRAM_AC26   | 53.4 | ✓ dram_ac_26-> [N1] DRAM_AC26       |
| A4                                                | DRAM_AC27   | 29   | ✓ dram_ac_27-> [R6] DRAM_AC27       |
| A4                                                | DRAM_AC28   | 31.9 | ✓ dram_ac_28-> [W6] DRAM_AC28       |
| A4                                                | DRAM_AC29   | 34.4 | ✓ dram_ac_29-> [V6] DRAM_AC29       |
| A4                                                | DRAM_AC30   | 59.5 | ✓ dram_ac_30-> [AC4] DRAM_AC30      |
| A5                                                | DRAM_AC31   | 56.5 | ✓ dram_ac_31-> [AD5] DRAM_AC31      |
| A4                                                | DRAM_AC32   | 34.3 | ✓ dram_ac_32-> [R4] DRAM_AC32       |
| A4                                                | DRAM_AC33   | 38.6 | ✓ dram_ac_33-> [R5] DRAM_AC33       |
| A4                                                | DRAM_AC34   | 43.1 | ✓ dram_ac_34-> [T1] DRAM_AC34       |
| A4                                                | DRAM_AC35   | 51.2 | ✓ dram_ac_35-> [T2] DRAM_AC35       |
| A4                                                | DRAM_AC36   | 26.5 | ✓ dram_ac_36-> [V5] DRAM_AC36       |
| A4                                                | DRAM_AC37   | 35.1 | ✓ dram_ac_37-> [W5] DRAM_AC37       |
| A4                                                | DRAM_AC38   | 42.1 | ✓ dram_ac_38-> [AB6] DRAM_AC38      |

|                          |                           |                  |
|--------------------------|---------------------------|------------------|
| Project M8 Minima.PrjPcb |                           | crls<br>design   |
| File 04-2 m8 io 1.SchDoc |                           |                  |
| Variant [No Variations]  | Page 6 / 16               |                  |
| Date 07-Sep-2026         | Revision                  | Unknown revision |
| Author Charles Surianto  | Reviewer Charles Surianto | 07-Sep-2026      |



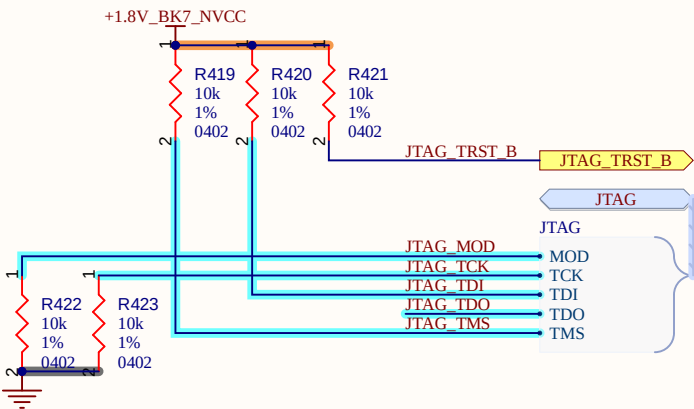
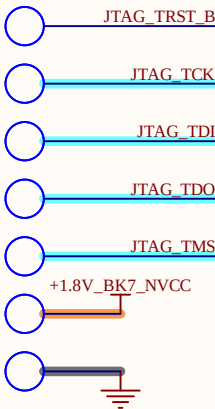
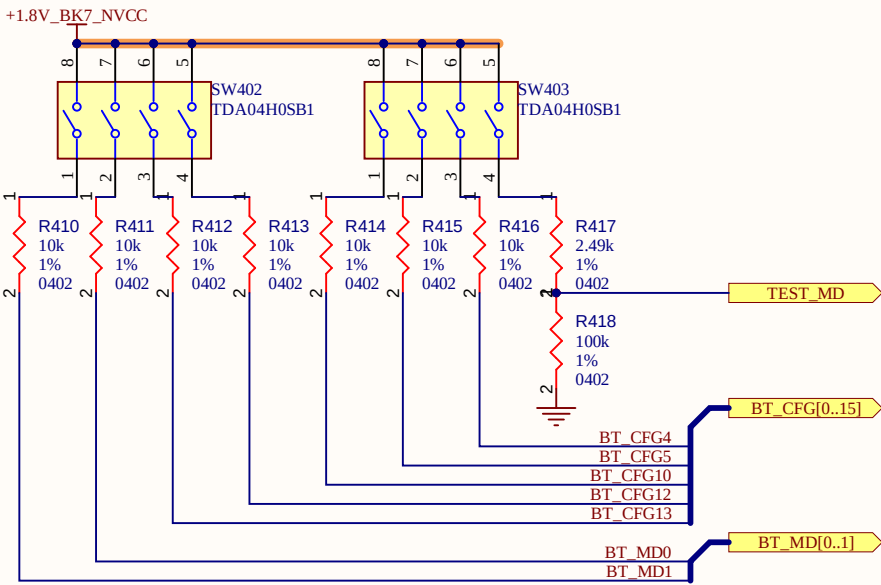
| Table 6-1. Boot MODE pin settings |                   |
|-----------------------------------|-------------------|
| BOOT_MODE[1:0]                    | Boot Type         |
| 00                                | Boot From Fuses   |
| 01                                | Serial Downloader |
| 10                                | Internal Boot     |
| 11                                | Reserved          |

| Table 6-9. Boot device selection |                             |
|----------------------------------|-----------------------------|
| BOOT_CFG[14:12]                  | Boot device                 |
| 001                              | SD/eSD                      |
| 010                              | MMC/eMMC                    |
| 011                              | NAND                        |
| 100                              | Serial NOR boot via FlexSPI |
| 110                              | Serial (SPI) NOR            |

| Table 6-42. SD/eSD Boot Fusemap |                                               |                    |   |                                          |                                                                                                                                     |   |                                                           |                                                                                                                   |
|---------------------------------|-----------------------------------------------|--------------------|---|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Addr                            | 7                                             | 6                  | 5 | 4                                        | 3                                                                                                                                   | 2 | 1                                                         | 0                                                                                                                 |
| 0x470[15:8]                     | Reserved                                      | Boot Device Select |   |                                          | Port Select:<br>00 - SD1<br>01 - SD2<br>10 - SD3                                                                                    |   | Power<br>Cycle<br>Enable<br><br>0 - Disable<br>1 - Enable | SD<br>Loopback<br>Clock<br>Source SEL<br>(SDR50 and<br>SDR104<br>Only)<br><br>0 - Through<br>SD pad<br>1 - Direct |
| 0x470[7:0]                      | Fast Boot:<br>0 - Regular<br>1 - Fast<br>Boot | Reserved           |   | Bus Width:<br><br>0 - 1-bit<br>1 - 4-bit | Speed<br><br>000 - Normal/SDR12<br>001 - High/SDR25<br>010 - SDR50<br>011 - SDR104<br>101 - Reserved for DDR50<br>Others - Reserved |   |                                                           | Reserved                                                                                                          |

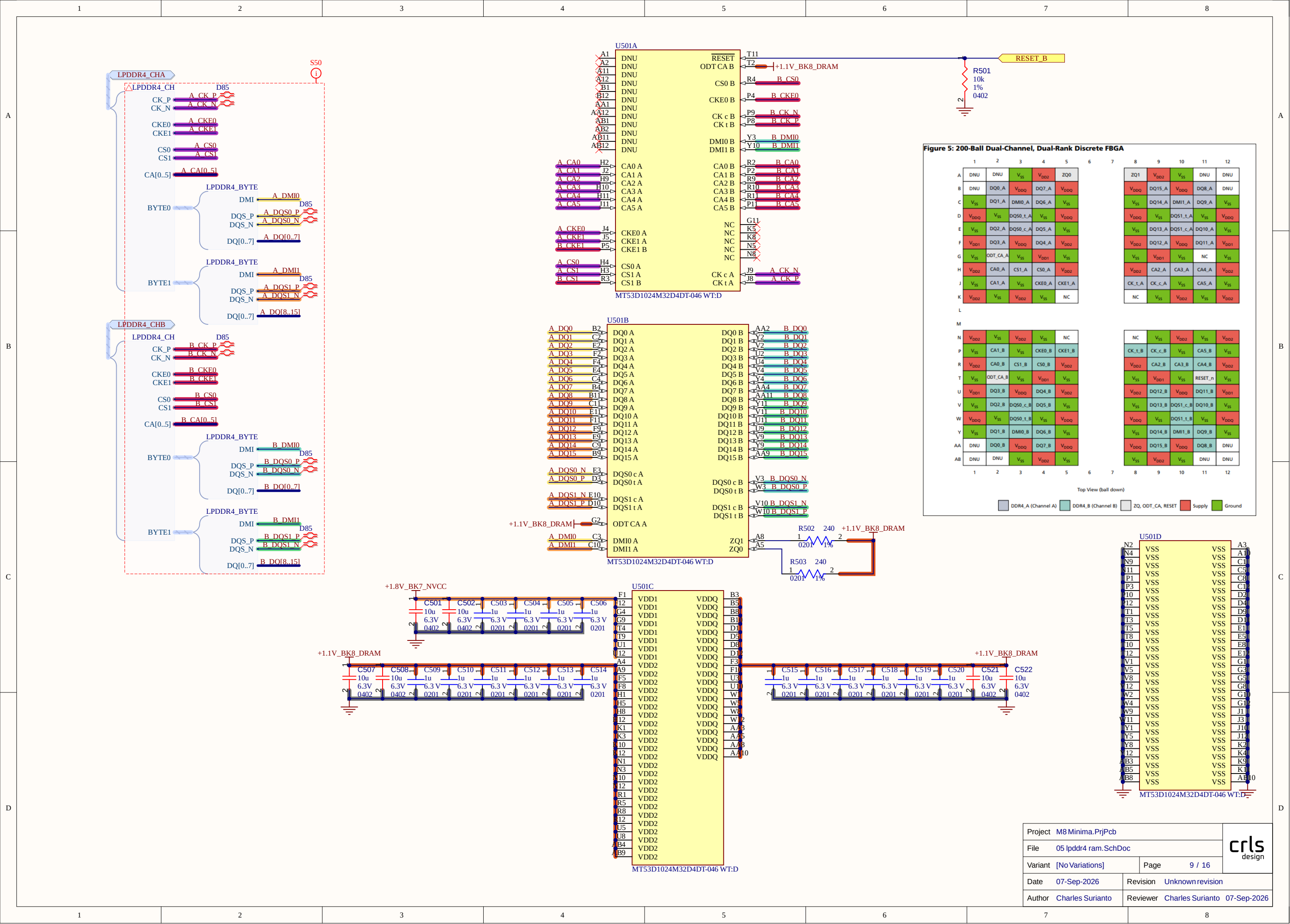
| Table 6-43. MMC/eMMC Boot Fusemap |                                            |                                                                                                                                      |   |   |                                                                               |   |                                                                            |                                                                                          |
|-----------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---|---|-------------------------------------------------------------------------------|---|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Addr                              | 7                                          | 6                                                                                                                                    | 5 | 4 | 3                                                                             | 2 | 1                                                                          | 0                                                                                        |
| 0x470[15:8]                       | Reserved                                   | Boot Device Select                                                                                                                   |   |   | Port Select:<br>00 - SD1<br>01 - SD2<br>10 - SD3                              |   | Power Cycle Enable<br>0 - Disable<br>1 - Enable                            | SD Loopback Clock Source SEL (SDR50 and SDR104 Only)<br>0 - Through SD pad<br>1 - Direct |
| 0x470[7:0]                        | Fast Boot:<br>0 - Regular<br>1 - Fast Boot | Bus Width:<br>000 - 1-bit<br>001 - 4-bit<br>010 - 8-bit<br>101 - 4-bit DDR (MMC 4.4)<br>110 - 8-bit DDR (MMC 4.4)<br>Else - Reserved |   |   | Speed<br>00 - Normal<br>01 - High<br>10 - Reserved for HS200<br>11 - Reserved |   | USDHC IO VOLTAGE SELECTION<br>For Normal Boot Mode<br>0 - 3.3V<br>1 - 1.8V | USDHC IO VOLTAGE SELECTION<br>For Manufacture Mode<br>0 - 3.3V<br>1 - 1.8V               |

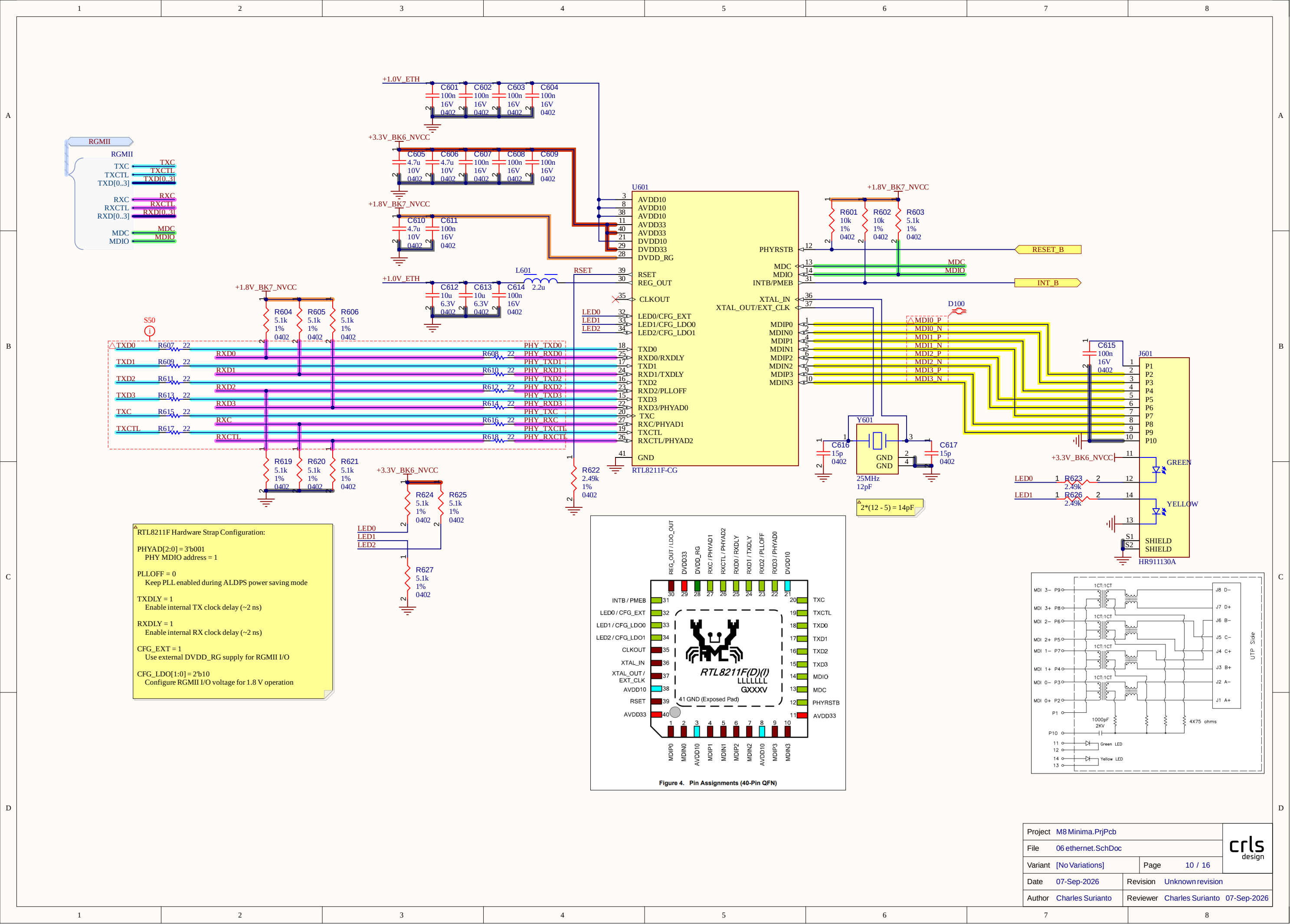
11|XXXXX|1 => JTAG  
01|XXXXX|0 => USB1 OTG  
10|10010|0 => SD1 eMMC 8bit  
10|011X1|0 => SD2 microSD 4bit



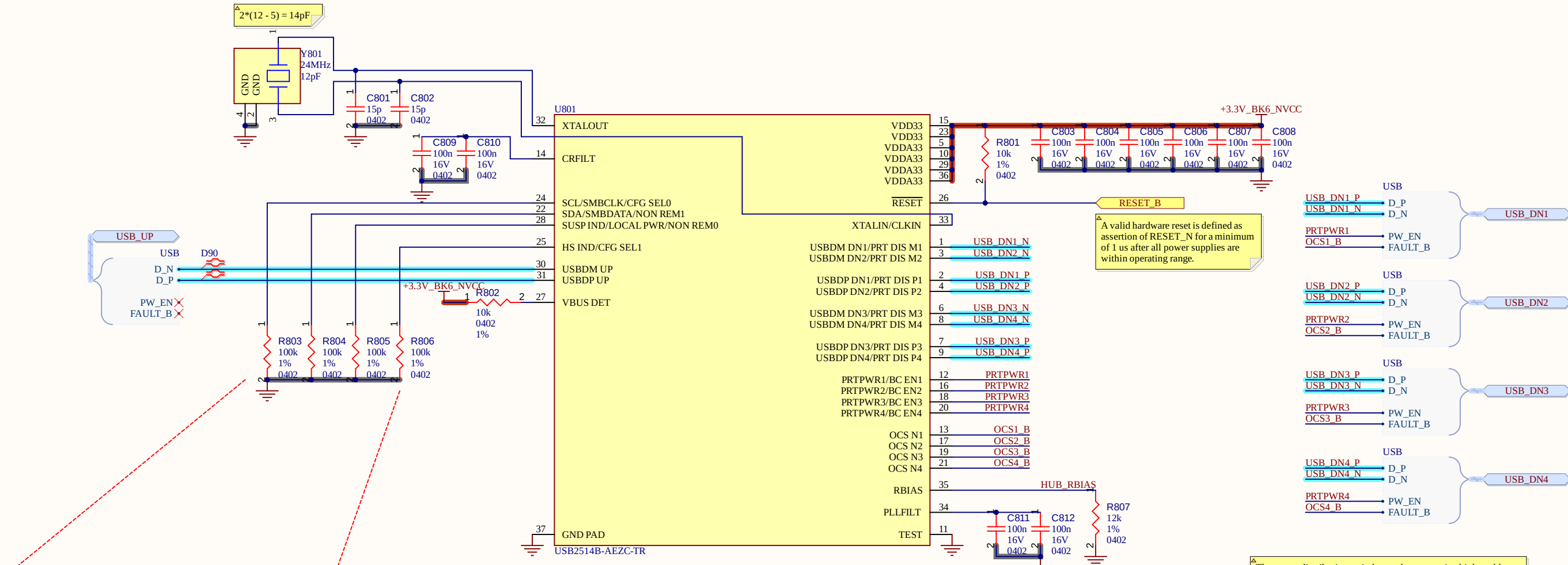
| Recommended board terminations for Boundary-scan |          |                      |                                                                                                 |
|--------------------------------------------------|----------|----------------------|-------------------------------------------------------------------------------------------------|
| Signal                                           | I/O type | External termination | Comments                                                                                        |
| BOOT_MODE0                                       | Input    | Pull-up              | BOOT_MODE0, BOOT_MODE1, JTAG_MOD, and TEST_MODE must be at 1101 to enter in Boundary-scan mode. |
| BOOT_MODE1                                       | Input    | Pull-up              |                                                                                                 |
| JTAG_MOD                                         | Input    | Pull-down            |                                                                                                 |
| TEST_MODE                                        | Input    | Pull-up              |                                                                                                 |
| GPIO1_IO02 (WDOG_B)                              | Input    | Pull-up              | External pull up resistor (100 Kilohms) must support boundary-scan mode.                        |





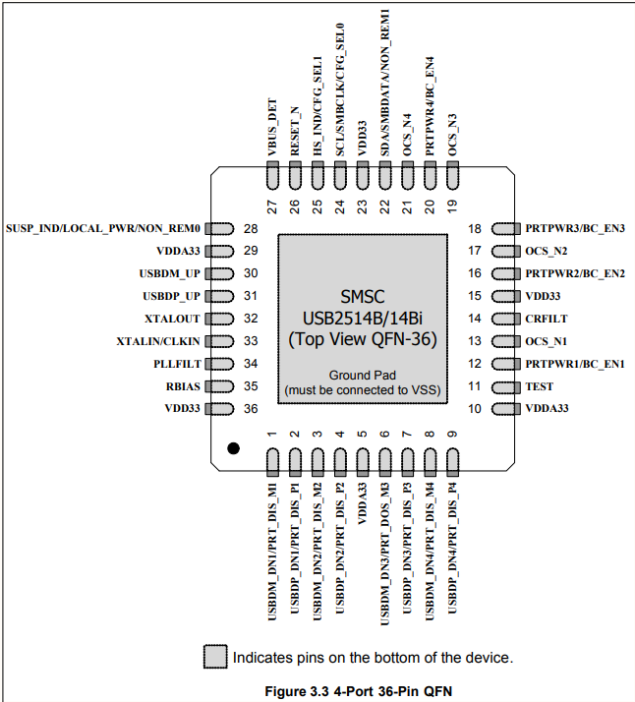




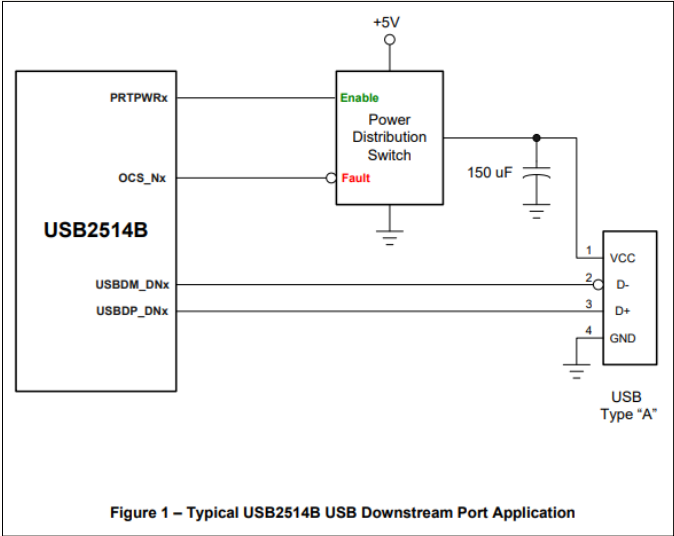


| Mode Configuration Straps |                                                                                                                                                     |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| CFG_SEL[1..0]             | Description                                                                                                                                         |
| 00                        | <b>Default Configuration</b><br>Strap Options Enabled, Self-Powered Operation Enabled, Individual Power Switching & Individual Over-Current Sensing |
| 01                        | <b>SMBus Slave Configuration</b><br>Strap Options Disabled, All Registers Configured Over SMBus                                                     |
| 10                        | <b>Default Configuration</b><br>Strap Options Enabled, Bus-Powered Operation Enabled, Individual Power Switching & Individual Over-Current Sensing  |
| 11                        | <b>I<sup>2</sup>C EEPROM Configuration</b><br>Strap Options Disabled, All Registers Configured By I <sup>2</sup> C EEPROM                           |

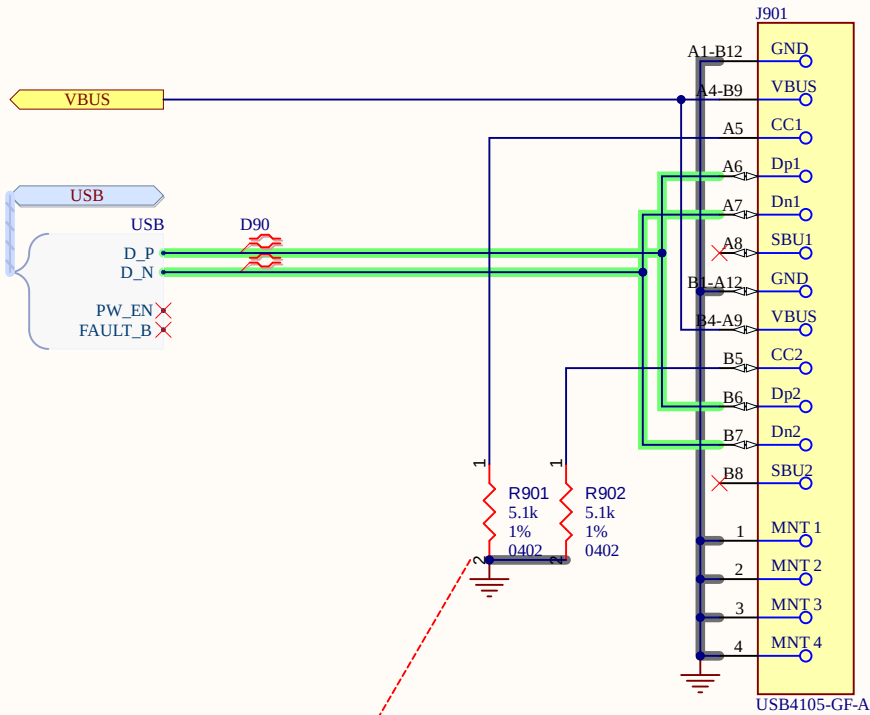
| NON_REMx Mode Configuration Straps |                                                                  |
|------------------------------------|------------------------------------------------------------------|
| NON_REM[1..0]                      | Description                                                      |
| 00                                 | All Downstream Ports Are Removable                               |
| 01                                 | Port 1 Is Non-Removable (Permanently Attached)                   |
| 10                                 | Port 1 & Port 2 Are Non-Removable (Permanently Attached)         |
| 11                                 | Port 1, Port 2 & Port 3 Are Non-Removable (Permanently Attached) |



A The power distribution switch must have an active high enable input and an over-current fault indication active low output.







|     |             |     |             |
|-----|-------------|-----|-------------|
| A1  | GND         | B12 | GND         |
| A4  | VBUS        | B9  | VBUS        |
| A5  | CC1         | B8  | SBU2        |
| A6  | DP1         | B7  | DN2         |
| A7  | DN1         | B6  | DP2         |
| A8  | SBU1        | B5  | CC2         |
| A9  | VBUS        | B4  | VBUS        |
| A12 | GND         | B1  | GND         |
| PIN | SIGNAL NAME | PIN | SIGNAL NAME |

TABLE 7: VALID UFP RD PULL-DOWN RESISTOR VALUES

| Rd Implementation     | Nominal Value | Detect Power Capability? | Current Source to 1.7V - 5.5V |
|-----------------------|---------------|--------------------------|-------------------------------|
| ± 20% voltage clamp   | 1.1V          | No                       | 1.32V                         |
| ± 20% resistor to GND | 5.1kΩ         | No                       | 2.18V                         |
| ± 10% resistor to GND | 5.1kΩ         | Yes                      | 2.04V                         |

A

B

C

D

A

B

C

D

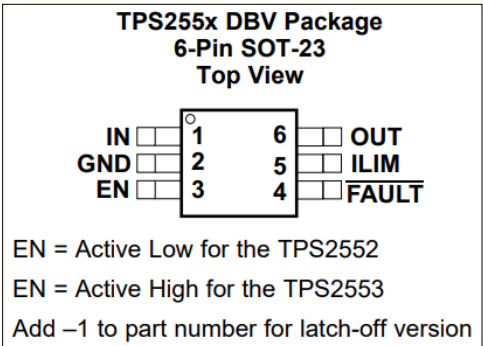
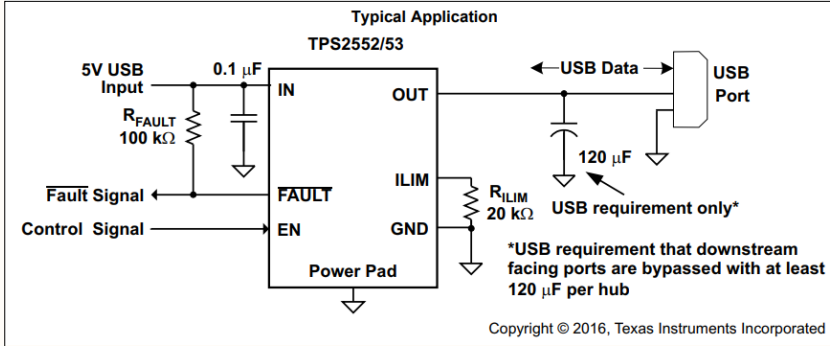
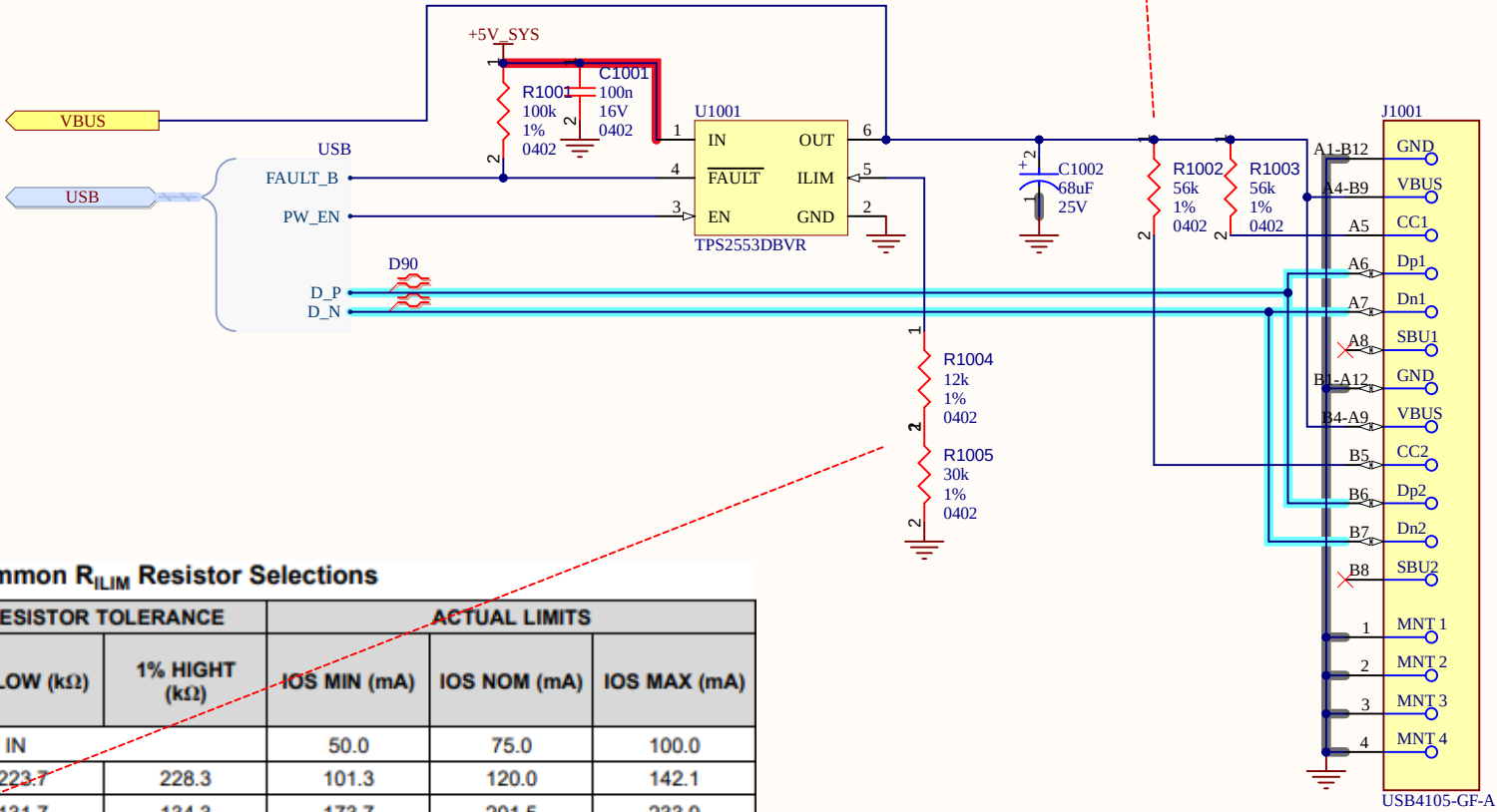


TABLE 6: VALID DFP RP PULL-UP RESISTOR VALUES

| DFP Current Capability                                 | Resistor Pull-up to 4.75V - 5.5V | Resistor Pull-up to 3.3V ± 5% | Current Source to 1.7V - 5.5V |
|--------------------------------------------------------|----------------------------------|-------------------------------|-------------------------------|
| Default USB Power (500mA for USB2.0, 900mA for USB3.0) | 56 kΩ ± 20%                      | 36 kΩ ± 20                    | 80 μA ± 20%                   |
| 1.5A @ 5V                                              | 22 kΩ ± 5%                       | 12 kΩ ± 5%                    | 180 μA ± 8%                   |
| 3.0A @ 5V                                              | 10 kΩ ± 5%                       | 4.7 kΩ ± 5%                   | 330 μA ± 8%                   |



|     |             |     |             |
|-----|-------------|-----|-------------|
| A1  | GND         | B12 | GND         |
| A4  | VBUS        | B9  | VBUS        |
| A5  | CC1         | B8  | SBU2        |
| A6  | DP1         | B7  | DN2         |
| A7  | DN1         | B6  | DP2         |
| A8  | SBU1        | B5  | CC2         |
| A9  | VBUS        | B4  | VBUS        |
| A12 | GND         | B1  | GND         |
| PIN | SIGNAL NAME | PIN | SIGNAL NAME |

Table 2. Common R<sub>ILIM</sub> Resistor Selections

| DESIRED<br>NOMINAL<br>CURRENT<br>LIMIT<br>(mA) | IDEAL<br>RESISTOR<br>(kΩ) | CLOSEST<br>1% RESISTOR<br>(kΩ) | RESISTOR TOLERANCE |                  | ACTUAL LIMITS |              |              |
|------------------------------------------------|---------------------------|--------------------------------|--------------------|------------------|---------------|--------------|--------------|
|                                                |                           |                                | 1% LOW (kΩ)        | 1% HIGHT<br>(kΩ) | IOS MIN (mA)  | IOS NOM (mA) | IOS MAX (mA) |
| 75                                             | SHORT ILIM to IN          |                                |                    |                  | 50.0          | 75.0         | 100.0        |
| 120                                            | 226.1                     | 226                            | 223.7              | 228.3            | 101.3         | 120.0        | 142.1        |
| 200                                            | 134.0                     | 133                            | 131.7              | 134.3            | 173.7         | 201.5        | 233.9        |
| 300                                            | 88.5                      | 88.7                           | 87.8               | 89.6             | 262.1         | 299.4        | 342.3        |
| 400                                            | 65.9                      | 66.5                           | 65.8               | 67.2             | 351.2         | 396.7        | 448.7        |
| 500                                            | 52.5                      | 52.3                           | 51.8               | 52.8             | 448.3         | 501.6        | 562.4        |
| 600                                            | 43.5                      | 43.2                           | 42.8               | 43.6             | 544.3         | 604.6        | 673.1        |
| 700                                            | 37.2                      | 37.4                           | 37.0               | 37.8             | 630.2         | 696.0        | 770.8        |
| 800                                            | 32.4                      | 32.4                           | 32.1               | 32.7             | 729.1         | 800.8        | 882.1        |
| 900                                            | 28.7                      | 28.7                           | 28.4               | 29.0             | 824.7         | 901.5        | 988.7        |
| 1000                                           | 25.8                      | 26.1                           | 25.8               | 26.4             | 908.3         | 989.1        | 1081.0       |
| 1100                                           | 23.4                      | 23.2                           | 23.0               | 23.4             | 1023.7        | 1109.7       | 1207.5       |
| 1200                                           | 21.4                      | 21.5                           | 21.3               | 21.7             | 1106.0        | 1195.4       | 1297.1       |
| 1300                                           | 19.7                      | 19.6                           | 19.4               | 19.8             | 1215.1        | 1308.5       | 1414.9       |
| 1400                                           | 18.3                      | 18.2                           | 18.0               | 18.4             | 1310.1        | 1406.7       | 1517.0       |
| 1500                                           | 17.0                      | 16.9                           | 16.7               | 17.1             | 1412.5        | 1512.4       | 1626.4       |
| 1600                                           | 16.0                      | 15.8                           | 15.6               | 16.0             | 1512.5        | 1615.2       | 1732.7       |
| 1700                                           | 15.0                      | 15.0                           | 14.9               | 15.2             | 1594.5        | 1699.3       | 1819.4       |

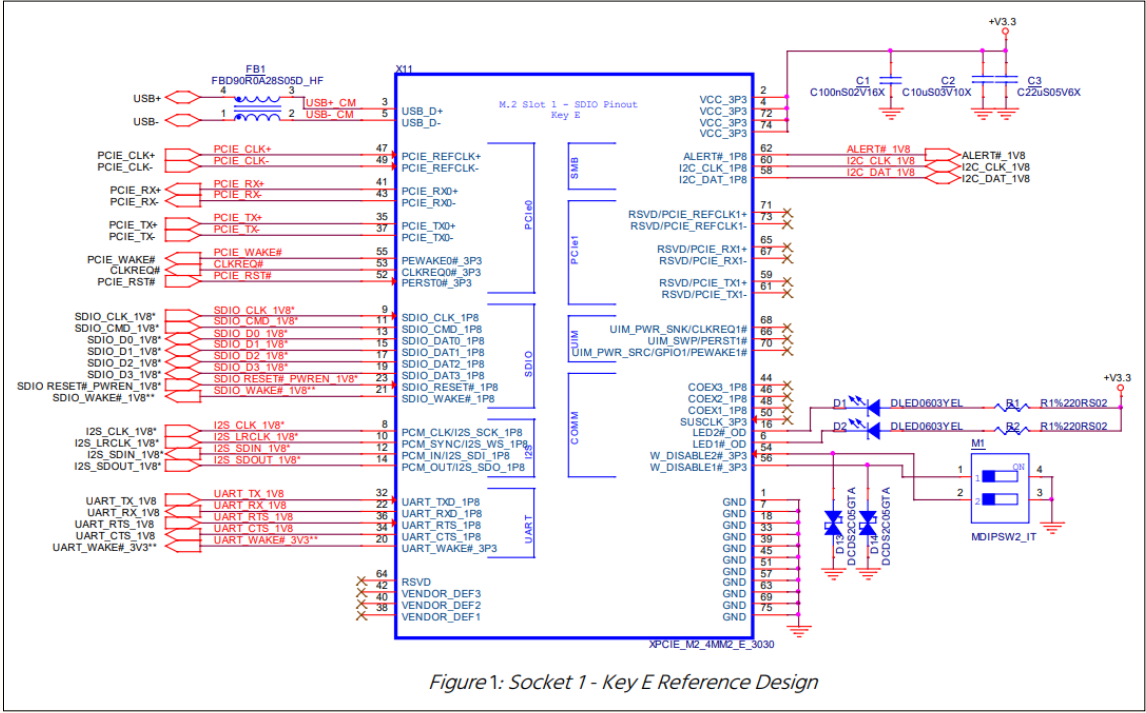


Figure 1: Socket 1 - Key E Reference Design

